MOHIT AGARWAL

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Roorkee, India | Remote, open to US time zones

SUMMARY

Mathematics and Computing student at IIT Roorkee and incoming Microsoft SDE intern. Builds backend and full-stack systems in TypeScript, Python and C++, with merged open-source contributions to WasmEdge and Oppia. Available for a remote internship from now through April 2027.

EDUCATION

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
B.S.-M.S. Dual Degree, Mathematics and Computing; CGPA 9.31/10 — *2023 – 2028 (expected)*
Incoming Software Development Engineer Intern, Microsoft (May 2027)
Relevant coursework: Data Structures and Algorithms, Database Management Systems, System Software, Fundamentals of AI/ML, Data Science, Introduction to Automata Theory, Discrete Mathematics

EXPERIENCE

Open Source Contributor, WasmEdge (CNCF WebAssembly runtime) — Jul 2026 – Aug 2026
- Developed a fix so the executor's atomic notify instruction uses the correct memory access width (merged).
- Implemented loader serializer tests for atomic.fence, wait64 and shared memory limits (merged).

Open Source Contributor, Oppia (open-source learning platform) — Feb 2025 – Mar 2025
- Developed fixes for dummy-topic generation on the admin page and a mismatch between topic preview and learner view (merged).
- Implemented clearer CI failure logging for full-stack tests and reverted a styling regression (merged).

PROJECTS

TradeLayer | *Solidity, TypeScript, Hedera testnet, Foundry, React* — Code | Demo | Sep 2026
- Built a confidential-trading prototype for ETHOnline 2026 using local enclave simulation and Alpaca paper trading. Browser-sealed orders, policy checks and pooled settlement; live CRE DON execution remained blocked by consensus.
- Implemented 4 TypeScript handlers for order intake, broker reconciliation, netted batch settlement and policy updates. Tested locally; isolated a CRE consensus failure caused by differing response headers and documented the limitation.
- Deployed 5 custom Solidity contracts, verified on Hedera testnet, alongside ERC-3643 equity tokens from Hedera's Asset Tokenization Studio (KYC, freeze, clearing) and HIP-1215 scheduled transactions for keeper-less refunds.
- Created 87 Foundry tests (12 prove isolation between institutions), 13 crypto test vectors shared by browser and enclave, and 15 gateway JWT checks.

GitCraft | *Node.js, Express, Next.js, Gemini and Claude APIs, MCP, GitHub webhooks* — Dec 2025
- Built a tool that keeps a repository's documentation (specs, ADRs, release notes) up to date in Craft through its MCP server; won 1st place at Craft Winter Challenge 2025.
- Integrated GitHub OAuth and webhooks so each push or PR triggers LLM analysis with Gemini or Claude and updates the docs.

Citadel: E-commerce Admin Dashboard | *Next.js, MongoDB, NextAuth, Zod* — Dec 2025 – Jan 2026
- Built a server-rendered Next.js admin dashboard with JWT auth, role-based access control, rate limiting and Zod-validated product workflows; deployed with API and security docs.

LLM Inference Tracing and Replay Platform | *C++17, llama.cpp, GGUF, FTXUI, CMake* — Code — Jun 2026
- Built C++17 telemetry with the llama.cpp graph-evaluation callback for layer timings and numerical anomaly logging. Added binary trace recording, model-free replay and an FTXUI terminal interface.

ACHIEVEMENTS

Reached **Codeforces Expert** (max 1644) and **CodeChef 3-star** (max 1713, global rank 55 in Starters 242).

Won **4 hackathons** (2 overall, 2 track prizes): 1st at Craft Winter Challenge 2025, VeChain Global Hackathon, 1st in the Oasis track at AKINDO Buildathon, and the UniDAO track at StackTooDeep 2.0.

Qualified as a **finalist in IMC Prosperity 4** (2026), ranking #138 of about 22,000 teams in Mission 1.

TECHNICAL SKILLS

Languages: Python, C++ (C++17), TypeScript, JavaScript, Solidity, SQL
Web and backend: React, Next.js, Node.js, Express, FastAPI, Django, REST APIs, MongoDB, Supabase, Firebase, Redis
Infra and tools: Docker, AWS, Azure, GitHub Actions (CI/CD), Nginx, Vercel, Render, CMake, Git, Claude Code, Cursor
AI and ML: LLM APIs (Gemini, Claude), MCP, AI agents (IQAI ADK-TS), llama.cpp, PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost, pandas, NumPy, OpenCV, SHAP